RTF alert-compression

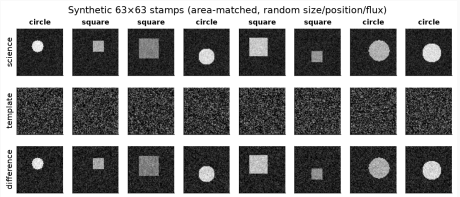
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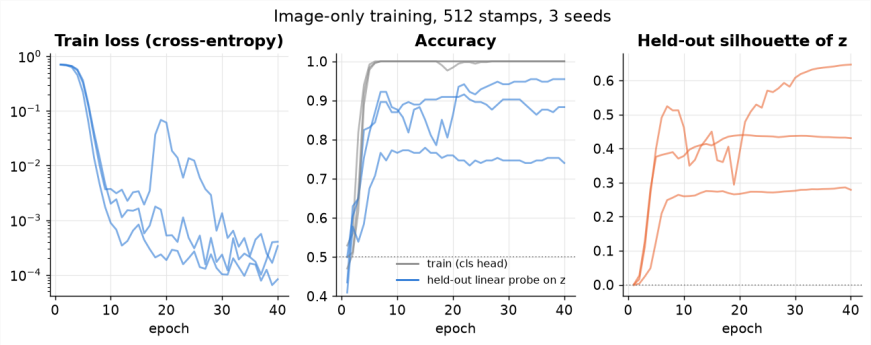
Question and setup

Does the `LightCurveCompressor` latent carry image information at all? Test it with no photometry, metadata or GP features.

- One identical all-zero photometry token per source, so the transformer CLS output is a constant (verified): $z = \mathrm{bottleneck}(\mathrm{const} + \mathrm{ImageTower}(\mathrm{stamp}))$.
- Stamps: 63×63 (science, template, difference); area-matched circles vs. squares with random size, position, flux and noise.
- Train image-only with cross-entropy on `cls_head`(z); score held-out latents with a linear probe, kNN and silhouette.
- Gotcha: `model.embed()` is `no_grad` and silently starves the image tower of gradient; train through `model.encode()`.

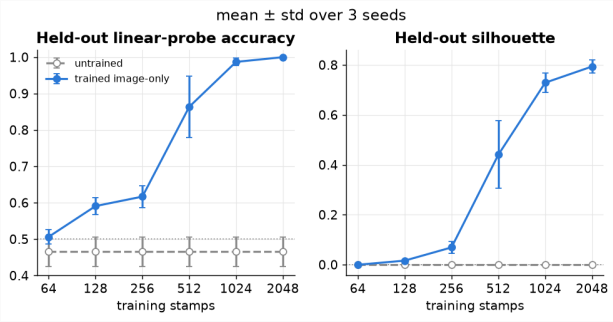


Yes: the latent separates circles from squares



- Training accuracy hits 100 % by epoch 6; held-out probe plateaus at 0.74–0.96 depending on seed. Thus, the gap is generalization, not necessarily optimization.
- 512 stamps, simple 0.25 M-parameter CNN, 30 epochs, ∼30 s on CPU.

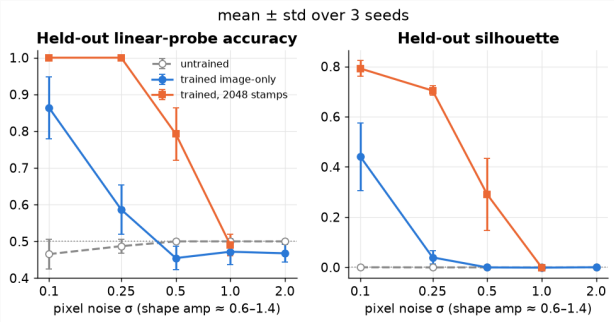
What the from-scratch CNN needs: data



training stamps	probe acc
64	0.51
256	0.62
512	0.86
1024	0.99
2048	1.00

Below ∼256 stamps nothing transferable is learned; the transition is between 256 and 1024.

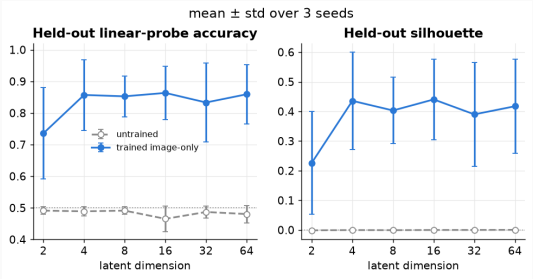
What the from-scratch CNN needs: signal, or more data

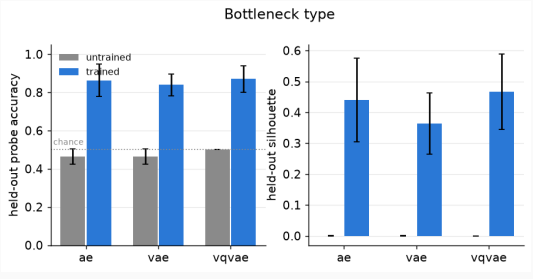


noise σ	512 st.	2048 st.
0.1	0.86	1.00
0.25	0.59	1.00
0.5	0.45	0.79
1.0	0.47	0.49

Blob brightness is 0.6–1.4. At 512 stamps the model collapses to chance by σ=0.5; with 2048 it is fine at 0.25 and useful at 0.5. Thus, the main issue is data.

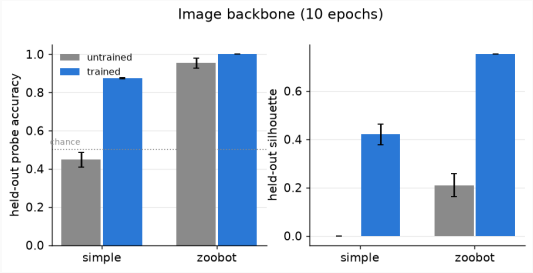
Things that do not matter: latent size, bottleneck type





- Latent dim 4–64: all 0.83–0.86, within seed variance. 2-d works on 2 of 3 seeds.
- AE 0.86, VAE 0.84, VQ-VAE 0.87. Two classes do not stress the bottleneck.

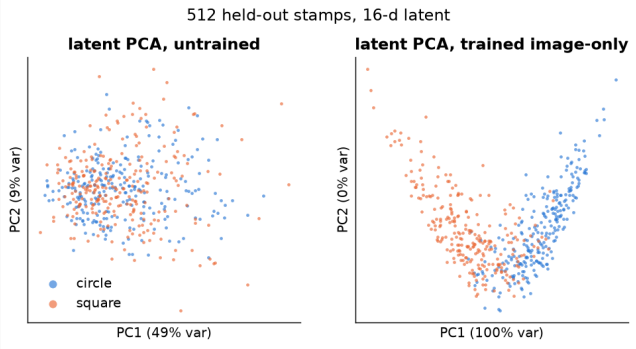
The thing that matters: the pretrained backbone



backbone	probe before	after
simple CNN (0.25 M)	0.45	0.87
Zoobot frozen (8.5 M)	**0.93–0.98**	**1.00**

Galaxy Zoo features separate the shapes linearly through a *random* projection, before any training. Training only the projection and bottleneck makes it perfect.

Latent geometry: the class loss collapses z to a line



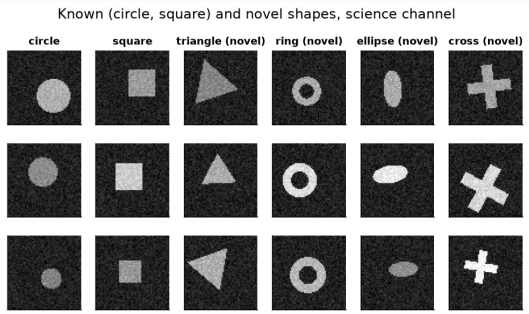
- After classification-only training, PC1 explains 100 % of the variance.
- Expected here: there is nothing else to encode. But it is the mechanism behind the next part.
- In the real model the photometry reconstruction keeps the other dimensions alive; it does not protect *image* information.

Part 2: can shapes the classifier never saw be found?

Train on circle/square only, then encode four unseen shapes: triangle, ring, ellipse, cross (area-matched, random orientation).

Read the latent without labels:

- novelty: 5-NN distance to known-class latents (AUROC)
- clustering: k-means, k=6 (AMI)
- ceiling: 6-way probe *with* labels
- geometry: participation ratio (16 = isotropic, 1 = line)



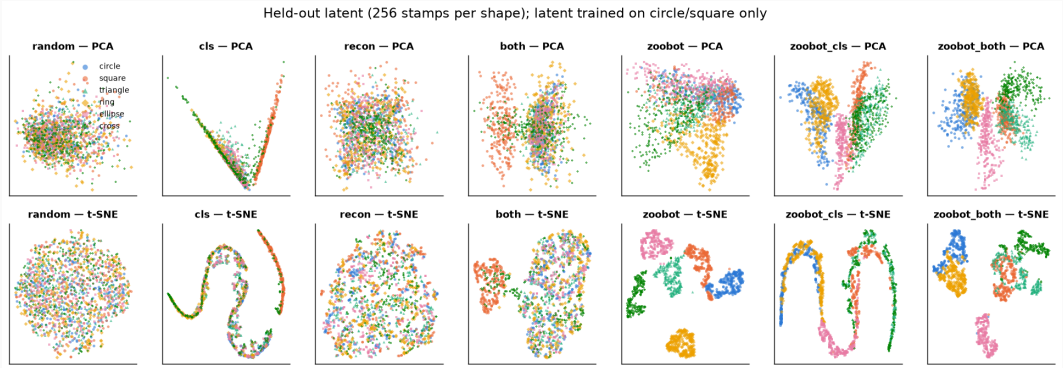
Seven latents: untrained CNN; CNN with class loss, stamp reconstruction, or both; Zoobot frozen with no training; Zoobot frozen with class loss or both. Zoobot is never fine-tuned; only projection, bottleneck and heads train.

Results across the seven latents

latent regime	probe (labels)	novelty AUROC	*k*-means AMI	eff. dims
untrained CNN	0.23	0.55	0.00	3.3
CNN, class loss	0.43	0.68	0.28	**1.0**
CNN, reconstruction	0.34	0.75	0.01	5.2
CNN, both	0.44	0.68	0.23	2.7
Zoobot frozen, untrained	**0.995**	**0.998**	0.54	4.8
Zoobot, class loss	0.85	0.96	0.54	**1.0**
Zoobot, both	0.97	0.97	**0.62**	1.4

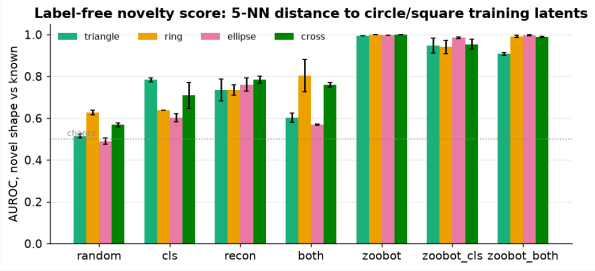
- No from-scratch CNN latent finds the novel shapes, whatever the loss.
- The frozen Zoobot latent finds all four with *zero* training.
- The class loss flattens every latent to one effective dimension, Zoobot included.

What the latents look like



Top: PCA (honest linear view). Bottom: t-SNE (exaggerates separation). Class-trained CNN: a V with novel shapes along the circle–square axis. Zoobot untrained: ring and cross clearly apart; triangle and ellipse overlap in PCA.

Label-free novelty and clustering



- Class-trained CNN calls 100 % of triangles, rings and ellipses "circle" with full confidence: classifier confidence is at chance as a novelty signal.
- Zoobot untrained: ring gets a pure cluster (95 % recall, 100 % purity); triangle and ellipse share one, which caps the AMI at 0.54.
- Zoobot with class loss: 1-d latent, probe drops $0.995 \rightarrow 0.85$, but each novel shape still owns its own stretch of the curve (AUROC 0.96).

What this does and does not prove

Does prove

- Pretrained Zoobot features retain information about classes never trained on; a CNN trained from scratch on two classes learns a two-class detector and nothing else.
- Discoverability lives in the backbone, not in the transformer, bottleneck or head.
- The joint classification loss is the main threat to it.

Does not prove (yet)

- That *Galaxy Zoo* pretraining specifically is the reason. Missing controls: ImageNet-pretrained ConvNeXt-pico and a random-init ConvNeXt-pico, both cheap.
- Anything about real stamps: hosts, PSF, faint transients. The toy flatters any pretrained encoder.
- That Zoobot is better at the *trained* task; with 2048 stamps the CNN also reached 1.00.

Takeaways and next steps

- The cutout pathway works and the latent is image-informative.
- From-scratch image tower: needs $\gtrsim$1000 stamps and clean signal, and discards everything but the trained classes.
- Frozen Zoobot: data-efficient, noise-tolerant on this toy, and keeps unseen structure findable without labels.
- Next: run the ImageNet and random-init ConvNeXt controls; then measure the class-loss collapse on real alerts with `eval_discovery.py`.

Code: `src/sanity_shapes.py`, `src/shape_experiments.py`, `src/shape_novelty.py`. Write-up with all figures and raw JSON: `docs/shape_baseline.md`. Tests: `tests/test_shapes.py`, `tests/test_novelty.py`.